

Beyond Markowitz: an Institutional Analysis of AI-Driven Portfolio Optimization

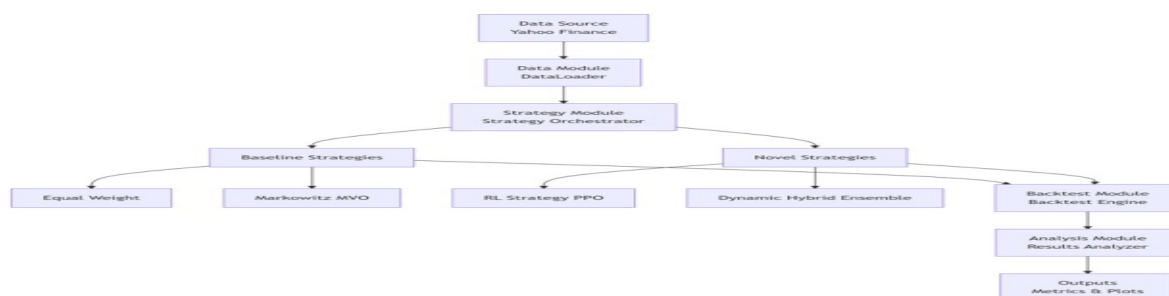
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Architectural Overview

High-level system architecture

The framework follows a modular, pipeline-based architecture as illustrated in the following diagram. The adopted approach ensures separation of concerns making the code easy to maintain and extend.



Technology Stack

- Programming language: Python 3.8+
- Key Libraries: pandas, numpy, yfinance, scikit-learn, stablebaselines3 (for PPO), pyportfoliopt(for calculating efficient frontier), matplotlib, plotly and seaborn

Functional Design

Core modules and dataflow: Data module, strategy module, backtest module, Analysis module

Novel Strategies

- Reinforcement learning (PPO) strategy
- Hybrid ensemble with dynamic switching

Deliverable

The final submission will be a GitHub repository whose structure will be as follows:

[README.md](#) and `final_project.ipynb`. `requirements.txt`, links to download data, source code and results are included in the notebook.

Risks and Assumptions

Risks

- RL instability- Since the PPO agent may fail to converge to a meaningful strategy, the fallback plan is to tune hyperparameters and use a simple, stable reward function
- data overfitting- the hybrid ensemble's dynamic switching mechanism could overfit to past market regimes. The mitigation measure adopted is to use walk-forward analysis and economically justified rules
- computational complexity- since training an RL agent is computationally expensive, the fallback plan is to use a small asset universe and limit historical data range

Assumptions

- The project relies on historical data as a reasonable proxy for future market behavior
- The transaction costs are modeled as a fixed percentage
- The selected asset universes are sufficiently liquid for the strategies implemented.